

Principal Component Analysis of the PEAK Relational Training System

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Abstract The purpose of the present study was to examine the underlying constructs within the Promoting the Emergence of Advanced Knowledge (PEAK) Relational Training System Direct Training Module. Ninety-eight children with Autism or other developmental and intellectual disabilities were evaluated using the PEAK Direct Training Assessment, which endorsed the presence or absence of 184 possible skills within their repertoire. Following this evaluation, a Principal Component Analysis was run which yielded a four-factor model of the PEAK Direct Training Module. Specifically, factors that possessed Eigenvalues greater than three were retained. Items were loaded onto each factor based on their correlation scores within a final rotated component structure matrix. The resulting four-factor model includes the constructs of Foundational Learning Skills, Perceptual Learning Skills, Verbal Comprehension Skills, and Verbal Reasoning, Memory, and Mathematical Skills. In an era of movement towards evidence-based practices, the present data provide support for PEAK system as a conceptually systematic approach to the treatment of children with autism.

Keywords ABA therapy · Language · Verbal behavior therapy

In recent years, there has been a steady rise in the number of individuals diagnosed with developmental disabilities. Between 1997 and 2008, Boyle et al. (2011) found a 17.1 % increase in the number of children aged three to eight with a diagnosed developmental disability, and a much larger increase of 289.5 % in autism diagnoses. Current rates of autism as estimated by the Center for Disease Control's Autism and Developmental Disabilities Monitoring Network Surveillance Year 2010 Principal Investigators (2014) indicate that 1 in 68 children have a diagnosis of autism spectrum disorder. This is a 30 % increase in the number of diagnoses since 2008. The increasing trend in the number of individuals diagnosed with developmental disabilities has led to an increase in the demand for treatment. The United States Department of Education has indicated

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